User Interface Design

Lab – 1

What is HTML?

- HTML stands for Hyper Text Markup Language
- HTML is the standard markup language for creating Web pages
- HTML describes the structure of a Web page
- HTML consists of a series of elements
- HTML elements tell the browser how to display the content
- HTML elements label pieces of content such as "this is a heading", "this is a paragraph", "this is a link", etc.

Definition of HTML Tags:

- The **<!DOCTYPE html>** declaration defines that this document is an HTML5 document
- The **<html>** element is the root element of an HTML page
- The **<head>** element contains meta information about the HTML page
- The **<title>** element specifies a title for the HTML page (which is shown in the browser's title bar or in the page's tab)

- The **<body>** element defines the document's body, and is a container for all the visible contents, such as headings, paragraphs, images, hyperlinks, tables, lists, etc.
- The **<h1>** element defines a large heading
- The **<p>** element defines a paragraph

HTML Page Structure:

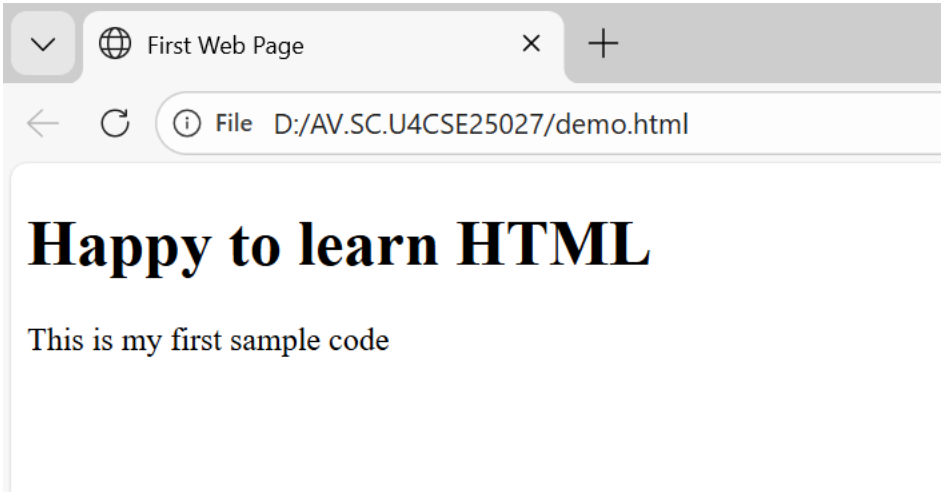
```
<html>
    <head>
        <title>Page title</title>
    </head>
    <body>
        <h1>This is a heading</h1>
        <p>This is a paragraph.</p>
        <p>This is another paragraph.</p>
    </body>
</html>
```

SAMPLE CODES:

1.Demo code:

```
<!DOCTYPE html>
```

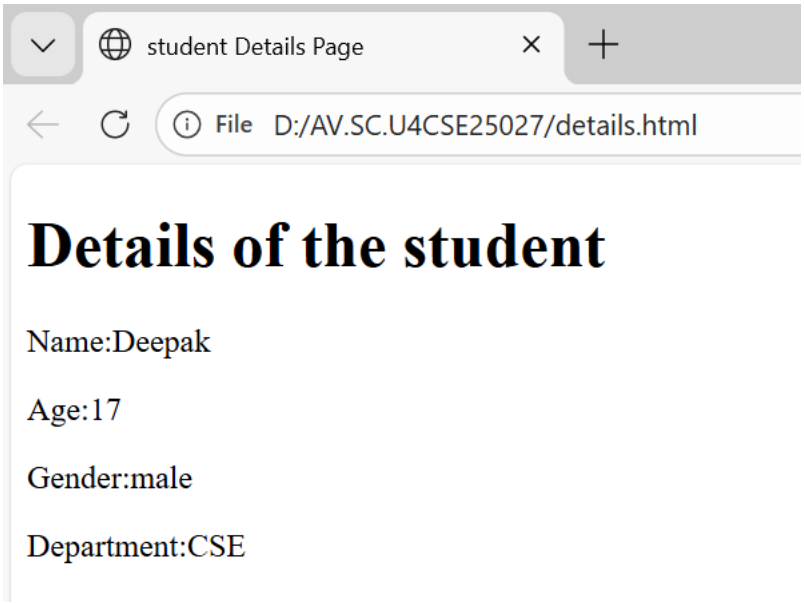
```
<html>
  <head>
    <title>First Web Page</title>
  </head>
  <body>
    <h1>Happy to learn HTML</h1>
    <p>This is my first sample code</p>
  </body>
</html>
```



2.Details of student:

```
<!DOCTYPE html>
<html>
  <head>
    <title>student Details Page</title>
  </head>
```

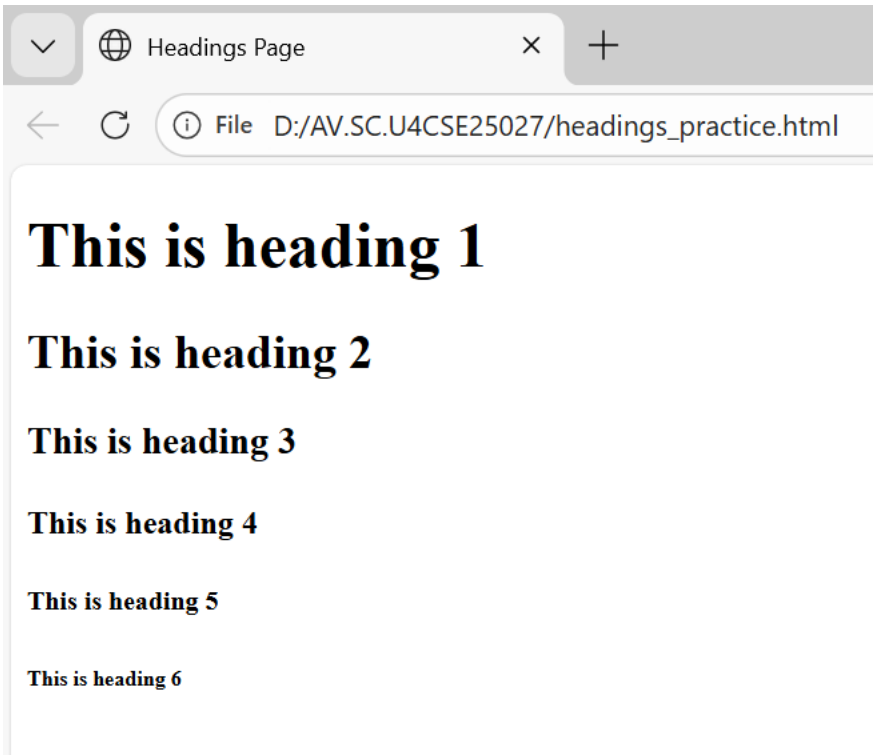
```
  <body>
    <h1>Details of the student</h1>
    <p>Name :Deepak</p>
    <p>Age :17</p>
    <p>Gender :male</p>
    <p>Department :CSE</p>
  </body>
</html>
```



3.Headings code:

```
<!DOCTYPE html>
<html>
  <head>
    <title>Details Page</title>
```

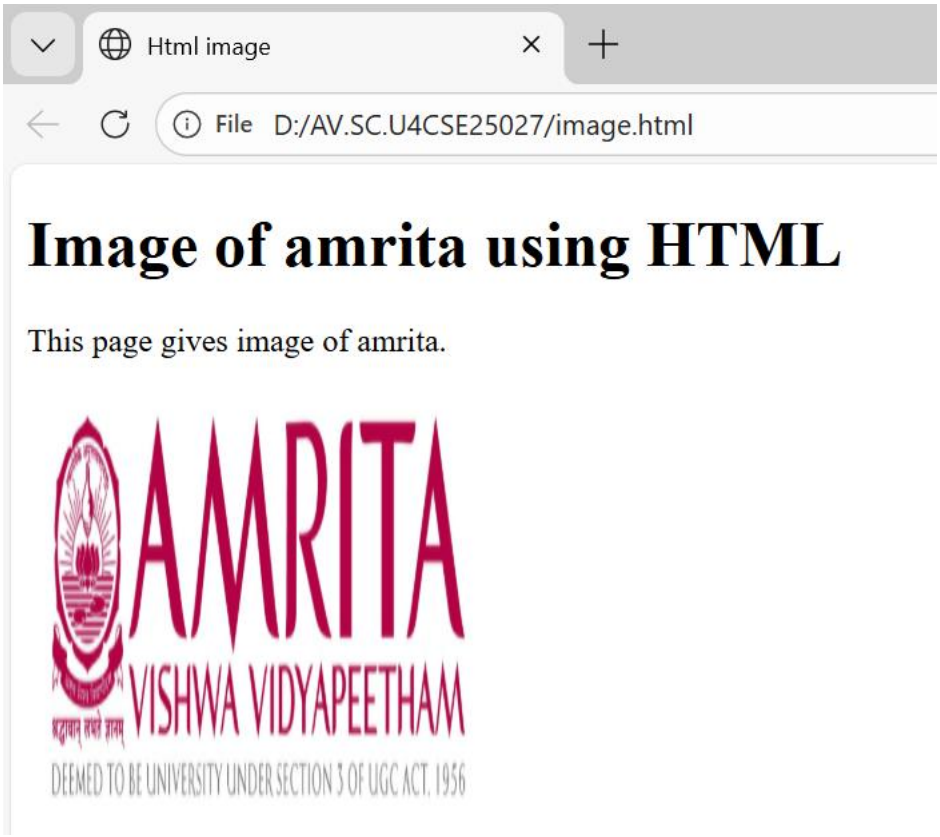
```
    </head>
    <body>
        <h1>This is heading 1</h1>
        <h2>This is heading 2</h2>
        <h3>This is heading 3</h3>
        <h4>This is heading 4</h4>
        <h5>This is heading 5</h5>
        <h6>This is heading 6</h6>
    </body>
</html>
```



4.Image code:

```
<!DOCTYPE html>
```

```
<html>
  <head>
    <title>Html image</title>
  </head>
  <body>
    <h1>Image of amrita using HTML</h1>
    <p>This page gives image of amrita.</p>
    <img src="C:\Users\DEEBAK\Downloads\amrita-vishwa-vidyapeetham-university-logo-2024.jpg" alt="amrita.edu" width="225" height="225">
  </body>
</html>
```



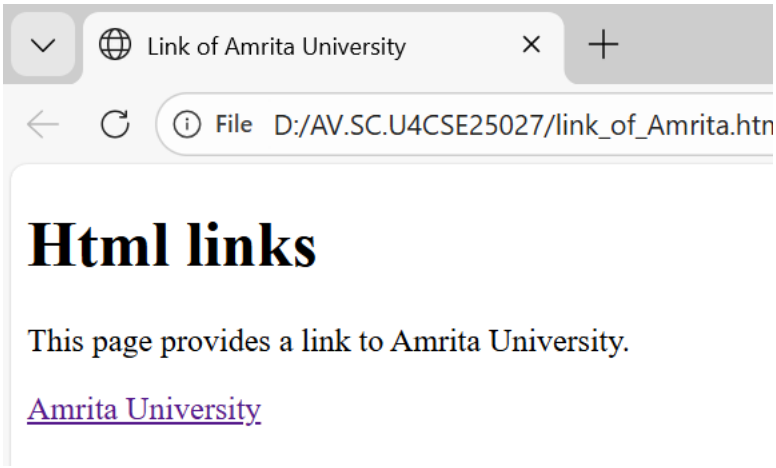
Why alt is used here:

The alt (alternative text) describes the image **in words**. It is shown or read **when the image cannot be displayed** and is also important for **accessibility**.

5.Links code:

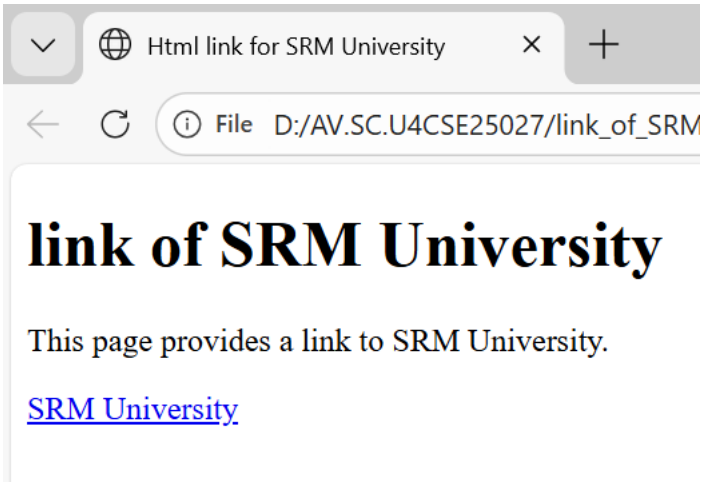
5.1 Amrita code:

```
<!DOCTYPE html>
<html>
  <head>
    <title>Link of Amrita University</title>
  </head>
  <body>
    <h1>Html links</h1>
    <p>This page provides a link to Amrita University.</p>
    <a href=https://www.amrita.edu/>Amrita University</a><br>
  </body>
```



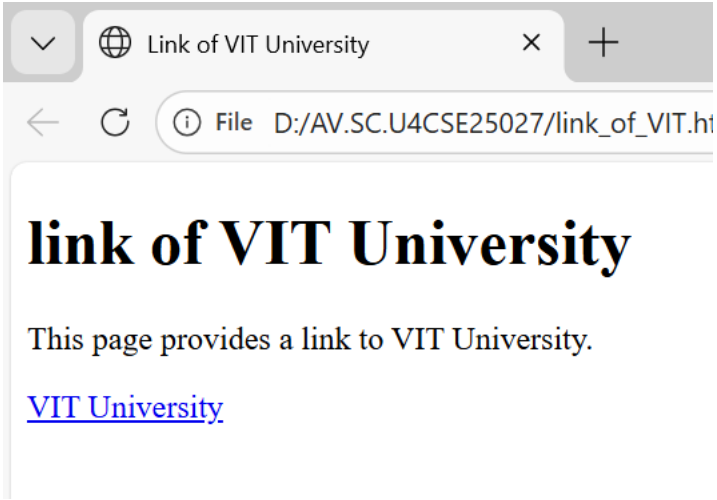
5.2 SRM University code:

```
<!DOCTYPE html>
<html>
  <head>
    <title>Html link for SRM University</title>
  </head>
  <body>
    <h1>link of SRM University</h1>
    <p>This page provides a link to SRM University.</p>
    <a href=https://www.srmist.edu.in/>SRM University</a><br>
  </body>
</html>
```



5.3 VIT University code:

```
<!DOCTYPE html>
<html>
  <head>
    <title>Link of Amrita University</title>
  </head>
  <body>
    <h1>link of VIT University</h1>
    <p>This page provides a link to VIT University.</p>
    <a href=https://www.vit.edu/>VIT University</a><br>
  </body>
</html>
```



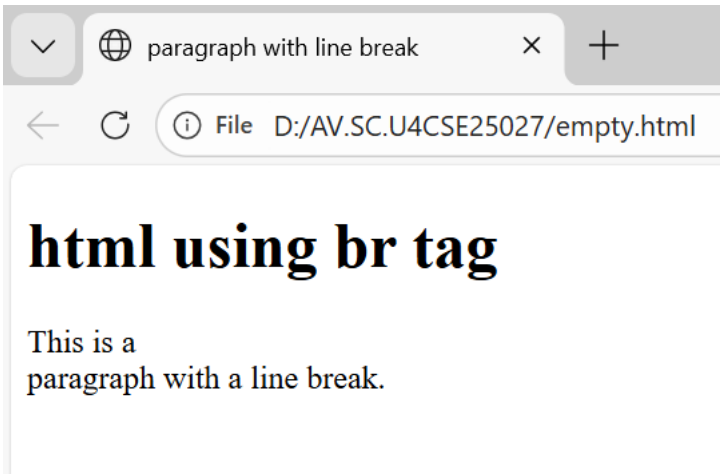
Why href is used here:

In HTML, the <a> (anchor) tag creates a **hyperlink**, but it does **nothing by itself** unless you tell it **where to go**.
The href (Hypertext Reference) attribute defines **the web address that opens when the link is clicked**.

6.Break using br code:

```
<!DOCTYPE html>
<html>
  <head>
    <title>paragraph with line break</title>
  </head>
  <body>
    <h1>html using br tag</h1>
    <p>This is a <br> paragraph with a line break.</p>
  </body>
```

```
</html>
```



Why did we use br tag:

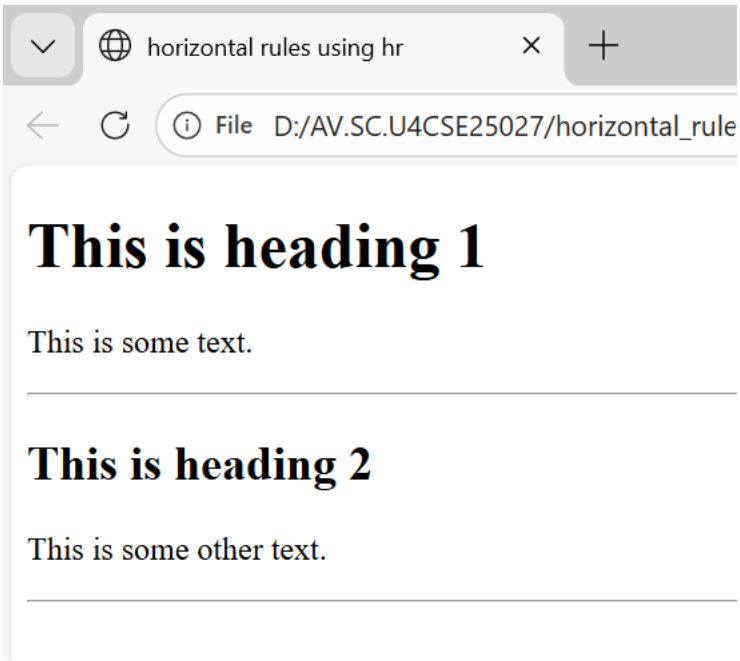
In HTML, text inside a <p> (paragraph) tag normally flows **continuously on the same line** until the browser automatically wraps it.
The <br> tag forces the text to **move to a new line without starting a new paragraph**.

 is an **empty tag**, so it does not need a closing tag.

7.Using hr:

```
<!DOCTYPE html>
<html>
  <head>
    <title>paragraph with line break</title>
  </head>
  <body>
    <h1>html using hr tag</h1>
```

```
    <p>This is a <br> paragraph with a line break.</p>
  </body>
</html>
```



Why did I use hr tag:

In HTML, the <hr> tag represents a **thematic break** between different sections of content. It helps the reader understand that one topic or section has ended and another has begun.

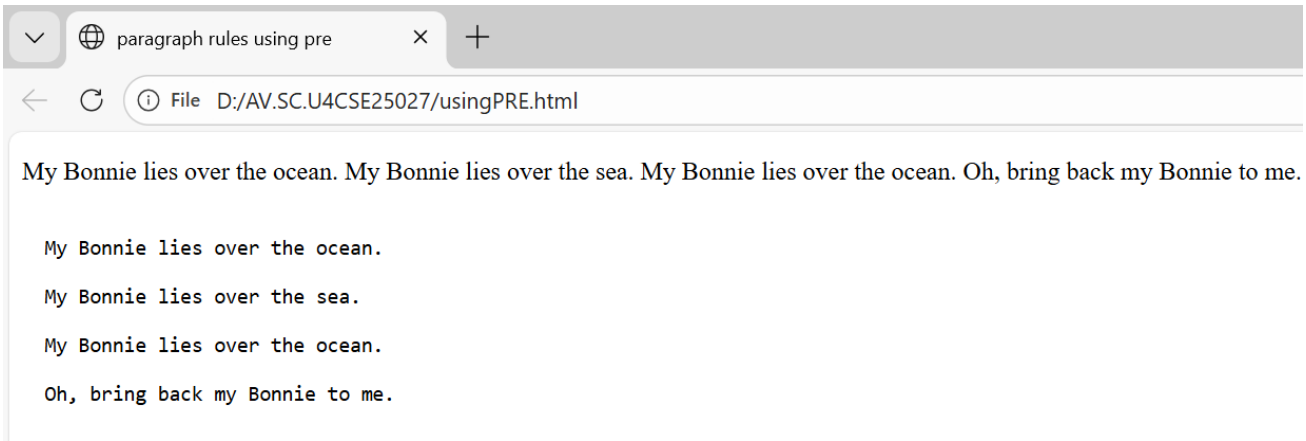
<hr> is an **empty tag**, so it does not need a closing tag.

8.using pre code:

```
<!DOCTYPE html>
<html>
  <head>
    <title>horizontal rules using hr</title>
  </head>
```

```
<body>
    <p>
My Bonnie lies over the ocean.

My Bonnie lies over the sea.

My Bonnie lies over the ocean.

Oh, bring back my Bonnie to me.

    </p>

    <pre>
My Bonnie lies over the ocean.

My Bonnie lies over the sea.

My Bonnie lies over the ocean.

Oh, bring back my Bonnie to me.
  </pre>
  </body>
```

</html>



Why did we use pre tag:

In normal HTML paragraphs (<p>), the browser **collapses extra spaces and line breaks**.
That's why the formatting inside the first <p> tag does **not** appear the same as how you typed it in the code.

The <pre> tag (preformatted text) tells the browser:

"Show this text exactly as written."